Sentiment Analysis-Based Study on Chinese Audiences Acceptance of "Friends"

Group 8 STAT 2630SEF

Github Repository

Introduction

Friends is a long-running sitcom that has attracted a large global audience, but cultural differences, such as plot developments, determine its acceptance in China.

This study uses a model to analyze online comments and explore the acceptance of Friends among Chinese audiences. By analyzing the emotions conveyed in the comments and key dialogue themes and phrases, it reveals some of the attitudes of Chinese viewers towards Friends.

System Design

The system is designed as a cloud-based data processing and analysis pipeline that integrates data storage, data processing, machine learning models, and visualization tools. The purpose of this system is to efficiently analyze user comments and generate meaningful insights through structured data processing and visualization.

The system begins with the data storage stage, where user comment data is stored in Azure Blob Storage. This cloud storage solution provides reliable and scalable storage for approximately 2,000 to 2,500 user comments collected from external sources.

Next, the data is transferred to MongoDB Atlas, which serves as the primary database system.

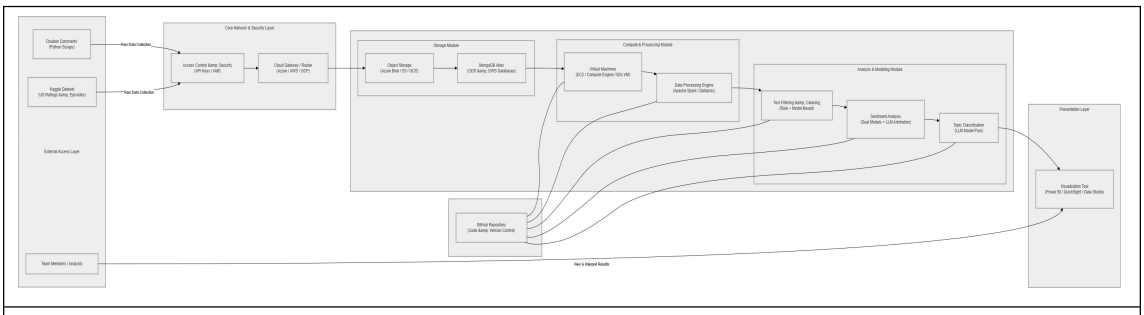
MongoDB Atlas is used to organize and manage structured datasets, allowing efficient data retrieval and processing during the analysis stage.

The data processing stage is implemented using Apache Spark, which supports large-scale data processing and improves the efficiency of data cleaning and transformation tasks. During this stage, Python is used as the primary programming language within the Google Colab cloud platform. Python scripts are applied to clean the dataset, remove duplicate records, and prepare the data for machine learning analysis.

After data processing, the system performs machine learning analysis using classification models such as Logistic Regression and Naive Bayes. These models are applied to perform sentiment analysis and categorize user comments into different emotional groups. The use of multiple models improves the reliability and accuracy of the classification results.

Finally, the processed results are exported to Microsoft Power BI, which is used as the visualization tool. Power BI generates graphical dashboards, including line charts, bar charts, pie charts, and word clouds, allowing users to easily interpret trends and patterns in user opinions.

Overall, this system design provides an efficient and scalable workflow that supports data storage, processing, analysis, and visualization in an integrated cloud-based environment.



System Architecture

Budget Plan

To evaluate the feasibility of the proposed system, three vendors were considered for development. The estimated costs included computing reources, database services, data processing, stroage, and visualization tools. All configurations are designed to support he processing of approximately 2,000-2,500 user comments.

Vendor A – Microsoft Azure

Microsoft Azure is selected as the primary cloud platform due to its strong integration with data storage and visualization tools.

Component	Service	Monthly Cost(HKD)
Virtual Machine	Azure B2s VM(2vCPU,4GB RAM)	800
Database	MongoDB Atlas	600
Data Processing	Azure Spark	800
Storage	Azure Blob Storage	200
Visualization	Power BI	100

Total Monthly Cost:HKD2,700

Total Yearly Cost:HKD32,400

The total cost is well below the HKD 200,000 budget limit and is sufficient to process approximately 2,285 user comments.

Vendor B – Amazon AWS

Component	Service	Monthly Cost(HKD)
Virtual Machine	Amazon EC2 t3.medium	900
Database	AWS	600
Data Processing	Spark	900
Storage	Amazon S3	250
Visualization	AWS QucikSight	150

Total Monthly Cost:HKD2,800

Total Yearly Cost:HKD33,600

The AWS-based system provides similar computing capability and supports large-scale data processing using Spark. The total yearly cost remains significantly below the HKD 200,000 budget limit.

Vendor C – Google Cloud

Component	Service	Monthly Cost(HKD)
Virtual Machine	Google Compute Engine	850
Database	MongoDB Atlas	600
Data Processing	Google Dataproc	850
Storage	Google Cloud Storage	200
Visualization	Google Data Studio	120

Total Monthly Cost:HKD2,620

Total Yearly Cost:HKD31,440

The GCP-based configuration offers a cost-effective solution with strong integration for cloud-based data processing and visualization tasks.

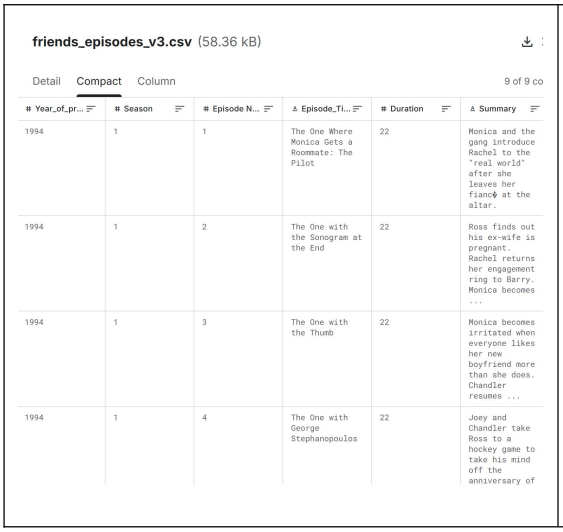
Method

We haved devided the whole into 4 steps, which are Data collect, Data Extract, Modeling, and Data Virtualization.

1. Data Collect (ODS->DWS)

1.1. Data Source

We use douban platform as our data source. Douban as most popular and television review community in China. We through this platform to scrapy chinese users short comments and rating. And then, in order to compare with the rating between China and America, we get some csv files from Kaggle.

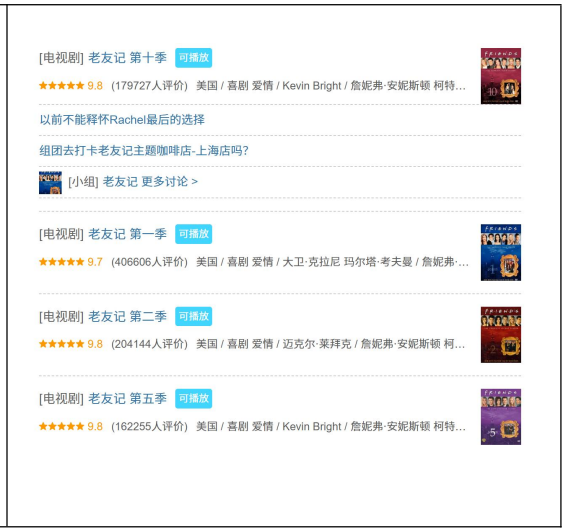


friends_episodes_v3.csv (58.36 kB)

Detail Compact Column 9 of 9 co

# Year_of_pr...	# Season	# Episode N...	A Episode_Ti...	# Duration	A Summary
1994	1	1	The One Where Monica Gets a Roommate: The Pilot	22	Monica and the gang introduce Rachel to the "real world" after she leaves her fianc◆ at the altar.
1994	1	2	The One with the Sonogram at the End	22	Ross finds out his ex-wife is pregnant. Rachel returns her engagement ring to Barry. Monica becomes ...
1994	1	3	The One with the Thumb	22	Monica becomes irritated when everyone likes her new boyfriend more than she does. Chandler resumes ...
1994	1	4	The One with George Stephanopoulos	22	Joey and Chandler take Ross to a hockey game to take his mind off the anniversary of

Kaggle data file



[电视剧] 老友记 第十季 可播放

9.8 (179727人评价) 美国 / 喜剧 爱情 / Kevin Bright / 詹妮弗·安妮斯顿 柯特...

以前不能释怀Rachel最后的选择

组团去打卡老友记主题咖啡店-上海店吗?

[小组] 老友记 更多讨论 >

[电视剧] 老友记 第一季 可播放

9.7 (406606人评价) 美国 / 喜剧 爱情 / 大卫·克拉尼 玛尔塔·考夫曼 / 詹妮弗·...

[电视剧] 老友记 第二季 可播放

9.8 (204144人评价) 美国 / 喜剧 爱情 / 迈克尔·莱拜克 / 詹妮弗·安妮斯顿 柯...

[电视剧] 老友记 第五季 可播放

9.8 (162255人评价) 美国 / 喜剧 爱情 / Kevin Bright / 詹妮弗·安妮斯顿 柯特...

Douban platform

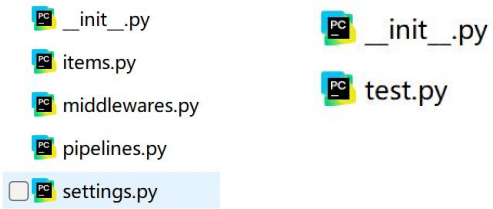
1.2. Acquisition Technology

We use Python scrapy to collect data from Douban platform. We used Python Scrapy as our crawling framework. The basic process of Python Scrapy have 4 steps.

Operation code	Examples of pictures

scrapy startproject+"the name of scrapy file"	`C:\Users\任钱\Desktop>scrapy startproject douban`
	`New Scrapy project 'douban', using template directory`
	`ckages\scrapy\templates\project', created in:`
	`C:\Users\任钱\Desktop\douban`
	
	`You can start your first spider with:`
	`cd douban`
	`scrapy genspider example example.com`

scrapy genspider+"scrapy name"+"crawler website"	`C:\Users\任钱\Desktop\douban>scrapy genspider test www.douban.com`
	`Created spider 'test' using template 'basic' in module:`
	`douban.spiders.test`

Python file edit	
scrapy crawl+"scrapy name"	`C:\Users\任钱\Desktop\douban>scrapy crawl test -o data.csv`

There also have more some detail about Python Scrapy.

Regarding 'Python file edit', the different Python file have different function. There we only talk some

basic files.

1) item.py: This file is used to set data schema. It defined data schema and data type so that provide uniform structural standard for data collect, data cleaning and storage.

2) settings.py: This file is used to set project configuration.

3) test.py: This is core crawler implementation file. You need to edit code to achieve crawl data from website to the specified file.

And then talk about scrapy output command.

scrapy crawl+"scrapy name": Look at picture, the '-o data.csv' represent that file name and the type of file. If you want to output json file, you can 'scrapy crawl test -o filename.json'.

1.3.Data Schema & Storage

1) Data Schema

About the data of douban platform, we design 2 tables, a 'ods_douban_comment table' and 'ods_douban_season_rating'. So they have different data schema.

Table 1: ods-douban-comment Fields: id season (int) comment (String) name (String)	Table 2: ods _ douban _ season _ rating Fields: season (string) rating (double) total _ people (int)

ods_douban _comment table ods_douban_season_rating table

Regarding the data that come from Kaggle, since they are already well organized and public accessible, so we preserved the orginal data structure and import data into a table that called 'ods_friends_episodes_v3'.

Table3: ods _ friends _ episodes _ v3
Fields: Year-of-prod (int)
Season (int)
Episode Number (int)
Episode _ Title (String)
Summary (String)
Director (String)
Rating (double)
Votes (int)

ods_friends_episodes_v3 table

2) Data Storage

We stored the collected data in the MongoDB-based ODS database. The full name of ODS is

'Operational Data Store'. This layer is used for temporary storage layer and construct a connection between raw data sources and data warehouse.

1.4.Data Size

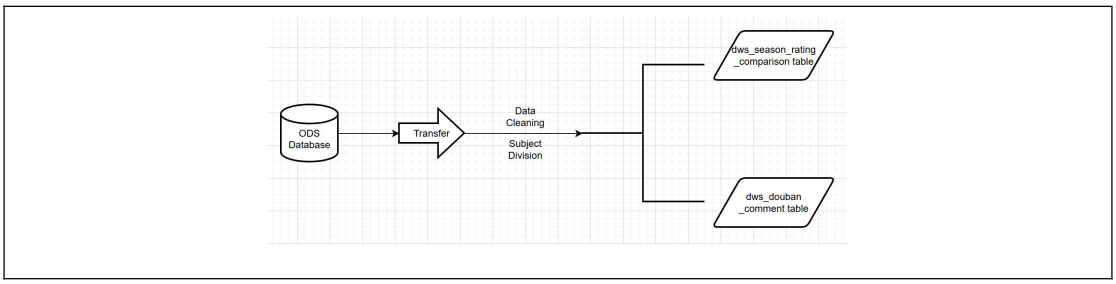
We use Python Scrapy to scrapy 4000 short comments from douban ,then output as CSV file. And we get 10 rating data of chinese audiences. Also we get 10 rating data of American audiences.

2. Data Extract

2.1. Process

Focus on Data Extract, we followed the path that from ODS to ETL process, and eventually to the DWS layer.

The full name of ETL is 'Extract, Transform, Laod'. The process that combines data from mutiple data sources into a single. In our project, we extracted some important data and loaded into MongoDB-based DWS database. We built this database through topic aggregation and data cleaning based on the ODS layer data.



The workflow of Data Extract

2.2. Detail

1) Data cleaning

In this project, during the ETL process from the ODS to the DWS layer, we first perform data deduplication, followed by sorting the records in ascending order based on the TV series season.

2) Subject divison

Based on the field topic of ODS layer, we divide and summarize the data into two core topic tables: rating comparison table and comment table. The first table aggregates rating indicators of each season to quantify audience feedback, and the second table aggregates audience comments of each season. After data cleaning and unifying the indicator caliber, we finally form a standardized and efficient indicator query service in DWS layer, providing solid data support for modeling and analyzing the acceptance of friends in China in ads layer.

3) Some code

In order to improve the code readability and data search effeciency, we use some sql statement.

```
#生成 dws_season_rating_comparison表
df_dws_rating=duckdb.query("""
  SELECT d.season,
      d.rating AS df_douban_rating,
      i.imdb_rating,
      ROUND((d.rating-i.imdb_rating),1) AS rating_difference
  FROM df_ods_douban_rating d
  LEFT JOIN (SELECT ROUND(AVG(Stars),1) AS imdb_rating,
             Season
         FROM df_ods_imdb_rating
         GROUP BY Season) i
  ON d.season=i.Season
  ORDER BY d.season
""").df()
```

```
#生成 dws_douban_comment
df_dws_comment=duckdb.query("""
  SELECT id AS user_id,
      season,
      "name" AS user_name,
      comment
  FROM df_ods_comment
  ORDER BY season
""").df()
```

2.3. Subject Table Construction

We constructed 2 subject table: a 'dws_season_rating_comparison' table and a 'dws_ douban_comment' table. They have different function.

1) dws_season_rating_comparison: Aggregates rating indicators of each season to quantify audience feedback.

2) dws_ douban_comment: Aggregates audience comments of each season.

Table 1. dws_season_rating_comparison Fields: season (int) douban-rating (double) imdb-rating (double) rating-difference (double)	Table 2: dws_douban_comment Fields: user_id name (String) comment (String) season (int)
dws_season_rating_comparison table	dws_douban_comment table

3. Modeling and Analysis Pipeline

The ADS layer is the core analysis module of this project. Its main function is to further transform the cleaned comment data in the DWS layer into structured analysis results that can be directly used for interpretation and visualization. In this project, the analysis process mainly revolves around two tasks: sentiment analysis and topic classification. The overall process consists of seven stages, including data filtering, text preprocessing, sentiment assessment, topic classification, result merging, and final result output. Through this process, the raw comment data can be gradually transformed into interpretable, statistically sound, and demonstrable analytical results.

3.1 Rule-based Filtering

The first stage employs a rule-based filtering method to remove low-quality comments before they enter the model analysis. This step mainly includes deleting empty or missing comments, standardizing text formatting, removing meaningless comments containing only symbols or numbers, and deleting duplicate records. This step serves as a preliminary data quality control step, reducing invalid input during subsequent model processing and minimizing unnecessary computational overhead, thus providing a cleaner corpus for subsequent analysis.

3.2 Model-based Filtering

After completing the rule-based filtering, the project further introduces a model-based filtering step to identify spam, offensive content, or comments lacking sufficient semantic information. Compared to relying on a single classifier, this stage employs a multi-model integrated approach to improve the

stability and reliability of the filtering results. Through this step, the retained comments are more semantically effective and better suited for subsequent sentiment analysis and topic classification tasks, thereby enhancing the overall credibility of the analysis results.

3.3 Text Pre-Processing

The filtered comments will then enter the text preprocessing stage to further improve the consistency and analysis ability of the text. This stage mainly includes Chinese word segmentation, stop word removal, and basic text normalization. Through this step, the comment text will be transformed into a form more suitable for model processing, providing higher-quality input data for subsequent sentiment analysis, keyword extraction, and topic classification.

3.4 Sentiment Analysis

Sentiment analysis is one of the core tasks of this project. To avoid instability issues caused by a single model, this project adopts a multi-model approach for sentiment assessment. Throughout the process, the system comprehensively uses different types of sentiment analysis models to jointly assess comment content. For samples with conflicting results or high uncertainty, additional judgment mechanisms are used to improve the stability and accuracy of the results. Ultimately, each comment is categorized into one of three sentiment categories: positive, neutral, or negative.

3.5 Topic Classification

Beyond assessing the sentiment of the comments, this project further categorizes the content into themes. Based on the project's analysis objectives, the comments are primarily divided into three core themes: acting, plot, and nostalgia. Through this stage, the system can not only determine whether viewers' attitudes towards *Friends* are positive or negative, but also further identify the key points of discussion, thus providing a more comprehensive explanation of the sources of audience acceptance.

3.6 Combining sentiment results and topic results

After completing sentiment analysis and topic classification, the system integrates the two results into a unified comment-level analysis table. Before merging, the topic classification results are deduplicated to ensure that each comment retains only one valid topic record. Then, the sentiment analysis results are used as the main component, and the topic classification results are merged in, so that each comment has both sentiment and topic tags. After this step, the data structure is more complete and facilitates subsequent statistical analysis and visualization.

3.7 Result Output

At the end of the analysis process, the system outputs two types of core result tables. The first type is a comment-level result table, which retains key information for each comment, including the quarter number, comment text, keywords, sentiment tags, and topic category. The second type is a quarterly summary result table, which summarizes the number of different sentiment categories by quarter and compares them with platform rating data. These two types of result tables together form the data foundation for subsequent Power BI visualization and final conclusion analysis.

4. Visualization

After completing the sentiment analysis and topic classification process, the results were presented through data visualization to provide a more intuitive understanding of audience perspectives and discussion trends. Based on the processed dataset, this project utilized Power BI to generate visualizations. This stage primarily focused on rating comparisons, topic distribution, sentiment distribution, and keyword frequency analysis, allowing users to efficiently interpret the analytical results.

The first category of visualization consists of a line chart, which compares ratings from different platforms. The results show that ratings on Douban are generally higher than those on IMDb. This comparison helps illustrate differences in audience perception between Chinese and international users,

indicating that Chinese audiences tend to give slightly higher ratings to the TV series *Friends*.

The second category of visualizations consists of bar charts, which illustrate the distribution of various discussion topics extracted from user comments. The results show that plot-related topics account for the largest number of comments, followed by nostalgia-related content, and then actor-related discussions. This indicates that audiences are primarily concerned with plot development, while emotional resonance and actor performance also play significant roles in audience engagement.

The third category of visualizations consists of pie charts, which present the distribution of sentiment categories across all comments. The charts show that the majority of comments express positive sentiment, while a smaller portion are neutral, and only a very small proportion are negative. This distribution reflects an overall positive attitude among audiences toward the series.

The final category of visualizations consists of a word cloud, which displays high-frequency keywords extracted from audience comments. Frequently appearing words such as “like,” “friends,” and “touching” highlight the emotional and interpersonal dimensions valued by viewers. These keywords suggest that themes such as friendship, emotional connection, and nostalgic memories have a strong influence on audience perceptions of the series.

Overall, the visualizations presented above provide clear and intuitive support for the analytical findings of this project. By transforming the processed textual data into graphical representations, the visualization stage enhances the readability of the results and effectively demonstrates that Friends continues to maintain a relatively high level of acceptance among Chinese audiences.

Result

In conclusion, through these visualizations, we can confidently conclude that Friends enjoys a high acceptance rate in China, characterized by high ratings, positive sentiment. According to graphs, we can find that words like' Classic, 'Like', and some character names appear largest. That also means Chinese viewers connect deeply with the characters and view the show as a classic.

Group Contribution

Hu Jihan : Make model for text analysis
Chen Jiahao: Data Collect and Data Extraction
Tang Chao: Data Visualization and System

GitHub Repository

Project GitHub Repository [*https://github.com/mgHurryo/STAT_2630SEF_Group_Project*]

References

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2. Kaggle
3. Hugging Face, for spam filtering, toxicity detection, sentiment analysis, and multilingual text embedding models used in this project.
4. GitHub, for open-source NLP libraries including jieba, KeyBERT, and SnowNLP; and stopwords-zh liabrary.
5. DeepSeek API, for LLM-based arbitration in sentiment analysis.
6. AliYun API, for LLM-based topic classification and model pool support.
7. Moonshot API, for supplementary LLM-based topic classification.